Fiscal Year:	FY 2017	**Task Last Updated:**	FY 06/28/2018
PI Name:	Nickerson, Cheryl A Ph.D.		
Project Title:	RNA Deep Sequencing and Metabolomic Profiling of Microgravity-Induced Regulation of the Host-Pathogen Interaction: An Integrated Systems Approach		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology		
Space Biology Cross-Element Discipline:	(1) Immunology		
Space Biology Special Category:	(1) Cell Culture (2) Translational (Countermeasure) Potential		
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PI Organization Type:	UNIVERSITY	**Phone:**	480-727-7520
Organization Name:	Arizona State University		
PI Address 1:	Center for Infectious Diseases and Vaccinology/The Biodesign Institute		
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City:	Tempe	**State:**	AZ
Zip Code:	85287-5401	**Congressional District:**	9
Comments:	NOTE PI moved from Tulane University to Arizona State University in 2006.		
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2012 Space Biology NNH12ZTT001N
Start Date:	07/01/2013	**End Date:**	09/30/2017
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
Contact Monitor:	Sato, Kevin	**Contact Phone:**	650-604-1104
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Flight Program:	ISS		
Flight Assignment:	NOTE: Extended to 9/30/2017 per NSSC information (Ed., 6/29/17) NOTE: Extended to 6/30/2017 per F. Hernandez/ARC (Ed., 4/14/16)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Aronow, Bruce Ph.D. (University of Cincinnati, College of Medicine) Khalpey, Zain M.D., Ph.D. (University of Arizona) Ott, Mark Ph.D. (NASA Johnson Space Center) Sams, Clarence Ph.D. (NASA Johnson Space Center) Barrila, Jennifer Ph.D. (CoPI/ Arizona State University)		
Grant/Contract No.:	NNX13AM01G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	A major challenge in mitigating infectious disease risks during spaceflight is to understand, model, and integrate the combined action of cellular, molecular, and biochemical networks in the host that potentiate transition to disease in response to infection by pathogens cultured in microgravity. This is a critical issue to address, as spaceflight negatively impacts crew immune function and alters microbial virulence, gene expression, and antimicrobial resistance, strongly suggesting an increased risk for in-flight infectious disease. Herein, we propose to apply our 3-D intestinal co-culture models as human surrogates to characterize their transcriptomic (via RNA-Seq), metabolomic (via mass spectrometry), and inflammatory responses to infection with spaceflight analogue and control-cultured Salmonella typhimurium. To facilitate a practical integration of these systems, the combined analysis proposed in this study will focus on the metabolomic subset associated with host oxidative stress, redox, and inflammatory responses. These factors are of key importance in the infection process and may contribute to reported abnormalities and dysfunction in the crews' immune system during flight. We will use systems modeling approaches based on both quantitative gene transcript and biochemical metabolite correlation and all of their known/inferable relationships between genes, proteins, pathways, and biochemistry that will allow us to form new bridges and a general framework for making these linkages between gene expression (RNA-Seq) and biochemical (metabolomic) data. This approach will allow for new insights into relationships between redox functions, oxidative stress, inflammation, and infection that either approach alone would not be able to achieve. By exploring interconnections between these systems over different kinetic timepoints before and after infection, this systematic approach will provide an unparalleled level of sensitivity and resolution of the dynamics of the host response to a microgravity-analogue cultured pathogen, which may lead to identification of novel infection mechanisms and strategies for prevention and control.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research will enrich life on Earth through the use of space technology and the application of biomedical knowledge. Specifically, this study will utilize the microgravity spaceflight platform to 1) to broaden our knowledge of the host-pathogen interaction that leads to infectious disease, and 2) for the development of new therapeutic strategies to combat infectious disease for the general public.
Task Progress:	Transcriptomic profiling of Salmonella-infected 3-D co-culture models has revealed distinct differences between low-shear modeled microgravity (LSMMG) and 1xg-infected cultures at select time points post-infection that supported infection trends from colonization experiments that had been performed in parallel. Transcriptomic differences between infected and time-matched uninfected cultures were observed at all time points tested. We also profiled transcriptomic differences between LSMMG and 1xg cultured bacteria alone (i.e., without host cells) immediately prior to infection of 3-D cultures to better help us explain potential host-pathogen interactions and colonization trends observed during this study. Transcriptomic, metabolomic, and cytokine data are currently being finalized and integrated using a multianalysis systematic approach for publication. Invited Presentations: Invited Speaker, C. Nickerson, Higher Orbits – Go for Launch! Space-inspired mission discovery STEM event for students grades 8-12. Mesquite High School, Gilbert, AZ, Feb 18-20, 2017 Invited Speaker, C. Nickerson, 3-D Mucosal Immune System Models for Health and Disease, presented at the “DECISIVE” New Projects in Synthetic Biology and Organism Engineering, Arizona State University, Feb 21, 2017 Invited Speaker, C. Nickerson, Outpacing Infectious Disease: Mimicking the host-pathogen microenvironment in vitro for studying mucosal infections, Center for Immunotherapy, Vaccines and Virotherapy. Arizona State University, Tempe, AZ, March 15, 2017 Invited Speaker, C. Nickerson, Regenerative Medicine/Tissue Engineering session for the 2nd Annual Molecular, Cellular, and Tissue Bioengineering Symposium, Arizona State University, April 1, 2017 Invited Speaker, J. Barrila, Using Spaceflight and Ground-based Modeled Microgravity Platforms for Novel Insight into Host-Pathogen Interactions. Center for Immunotherapy, Vaccines and Virotherapy. Arizona State University, Tempe, AZ, April 26, 2017 Barrila J, Yang J, Forsyth RJ, Gangaraju S, Ott CM, and Nickerson CA. Three-Dimensional Tissue Culture Models: Next Generation Predictive Preclinical Platforms for Human Health and Disease. Poster. Arizona Wellbeing Commons. Tempe, AZ (2017) Invited Speaker, C. Nickerson, American Society for Virology/American Society for Microbiology, Conference on Interplay of Viral and Bacterial Pathogens, May, 2017. *Jiseon Yang, postdoctoral fellow in my lab, presented this invited lecture.
Bibliography Type:	Description: (Last Updated: 07/02/2025)
Articles in Peer-reviewed Journals	Yang J, Barrila J, Roland KL, Ott CM, Nickerson CA. "Physiological fluid shear alters the virulence potential of invasive multidrug-resistant non-typhoidal Salmonella Typhimurium D23580." NPJ Microgravity. 2016 Jun 9;2:16021. eCollection 2016. https://doi.org/10.1038/npjmgrav.2016.21 ; PubMed PMID: 28725732; PubMed Central PMCID: PMC5515522 , Jun-2016
Articles in Peer-reviewed Journals	Barrila J, Yang J, Crabbé A, Sarker SF, Liu Y, Ott CM, Nelman-Gonzalez MA, Clemett SJ, Nydam SD, Forsyth RJ, Davis RR, Crucian BE, Quiriarte H, Roland KL, Brenneman K, Sams C, Loscher C, Nickerson CA. "Three-dimensional organotypic co-culture model of intestinal epithelial cells and macrophages to study Salmonella enterica colonization patterns." NPJ Microgravity. 2017 Feb 28;3:10. eCollection 2017. https://doi.org/10.1038/s41526-017-0011-2 ; PubMed PMID: 28649632; PubMed Central PMCID: PMC5460263 , Feb-2017
Articles in Peer-reviewed Journals	Barrila J, Crabbé A, Yang J, Franco K, Nydam SD, Forsyth RJ, Davis RR, Gangaraju S, Ott CM, Coyne CB, Bissell MJ, Nickerson CA. "Modeling host-pathogen interactions in the context of the microenvironment: 3-D cell culture comes of age." Infect Immun. 2018 Oct 25;86(11). pii: e00282-18. Print 2018 Nov. Review. https://doi.org/10.1128/IAI.00282-18 ; PubMed PMID: 30181350; PubMed Central PMCID: PMC6204695 , Oct-2018
Articles in Peer-reviewed Journals	Barrila J, Yang J, Franco Meléndez KP, Yang S, Buss K, Davis TJ, Aronow BJ, Bean HD, Davis RR, Forsyth RJ, Ott CM, Gangaraju S, Kang BY, Hanratty B, Nydam SD, Nauman EA, Kong W, Steel J, Nickerson CA. " Spaceflight analogue culture enhances the host-pathogen interaction between Salmonella and a 3-D biomimetic intestinal co-culture model." Front Cell Infect Microbiol. 2022 May 31;12:705647. https://doi.org/10.3389/fcimb.2022.705647 , May-2023
Awards	Nickerson CA, Pellis NR, Ott MC. (Editors) "2017 winner of the International Academy of Astronautics Life Science Book Award for 'Effect of Spaceflight and Spaceflight Analogue Culture on Human and Microbial Cells Novel Insights into Disease Mechanisms,' published by Springer, 2016." Jan-2017
Books/Book Chapters	Nickerson CA, Pellis NR, Ott CM. (Editors) "Effect of spaceflight and spaceflight analogue culture on human and microbial cells: Novel insights into disease mechanisms." Ed. C.A. Nickerson, N.R. Pellis, C.M. Ott. New York: Springer, 2016. 310 p. https://doi.org/10.1007/978-1-4939-3277-1 , Jun-2016
Books/Book Chapters	Nickerson CA, Ott CM. "Biomedical advances in three dimensions: An overview of human cellular studies in space and spaceflight analogues." in "Effect of spaceflight and spaceflight analogue culture on human and microbial cells: Novel insights into disease mechanisms." Ed. C.A. Nickerson, N.R. Pellis, C.M. Ott. New York: Springer, 2016. p. 83-92. https://doi.org/10.1007/978-1-4939-3277-1_4 , Jun-2016
Significant Media Coverage	Walter K. "3D Tissue Culture Model Mimics Gut Infections. Article on PI Nickerson's research, including quotes by her." R&D Magazine, March 10, 2017. https://www.rdmag.com/article/2017/03/3d-tissue-culture-model-mimics-gut-infections , Mar-2017
Significant Media Coverage	Arizona State University. "NASA, ASU Collaboration Develops 3-D Tissue Culture Models to Mimic Human Gut Infections. Article about PI Nickerson's research." Bioscience Technology online article, March 10, 2017. https://www.biosciencetechnology.com/news/2017/03/nasa-asu-collaboration-develops-3-d-tissue-culture-models-mimic-human-gut-infections , Mar-2017

Significant Media Coverage	Health Medicine Network. "NASA, ASU collaboration develops 3-D tissue culture models to mimic human gut infections. Onlnine article about PI Nickerson's research." Health Medicine Network, March 2017. http://healthmedicinet.com/i/nasa-asu-collaboration-develops-3-d-tissue-culture-models-to-mimic-human-gut-infections/ , Mar-2017
Significant Media Coverage	Arizona State University. "NASA, ASU collaboration develops 3-D tissue culture models to mimic human gut infections. Online article about PI Nickerson's research." EurekAlert! March 10, 2017. https://www.eurekalert.org/pub_releases/2017-03/asu-nac030917.php , Mar-2017
Significant Media Coverage	Arizona State University. "NASA, ASU Collaboration Develops 3-D Tissue Culture Models to Mimic Human Gut Infections. Online article about PI Nickerson's research." Infection Control Today, March 10, 2017. https://www.infectioncontroltoday.com/infections/nasa-asu-collaboration-develops-3-d-tissue-culture-models-mimic-human-gut-infections , Mar-2017
Significant Media Coverage	Scicasts staff. "Team Develops 3-D Tissue Culture Models to Mimic Human Gut Infections. Online article about PI Nickerson's research." SciCasts, March 13, 2017. https://scicasts.com/channels/disease-processes/1872-medical-microbiology/12372-team-develops-3-d-tissue-culture-models-to-mimic-human-gut-infections/ , Mar-2017